

INFORMATION TECHNOLOGY INSTITUTE

LAB 3

Cascode For Bandwidth





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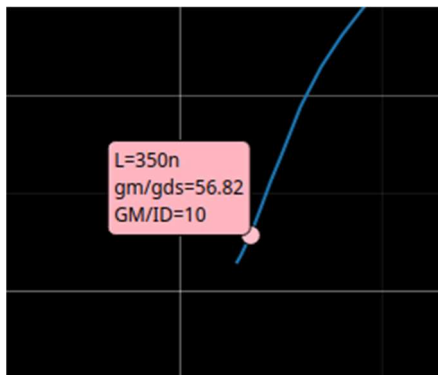
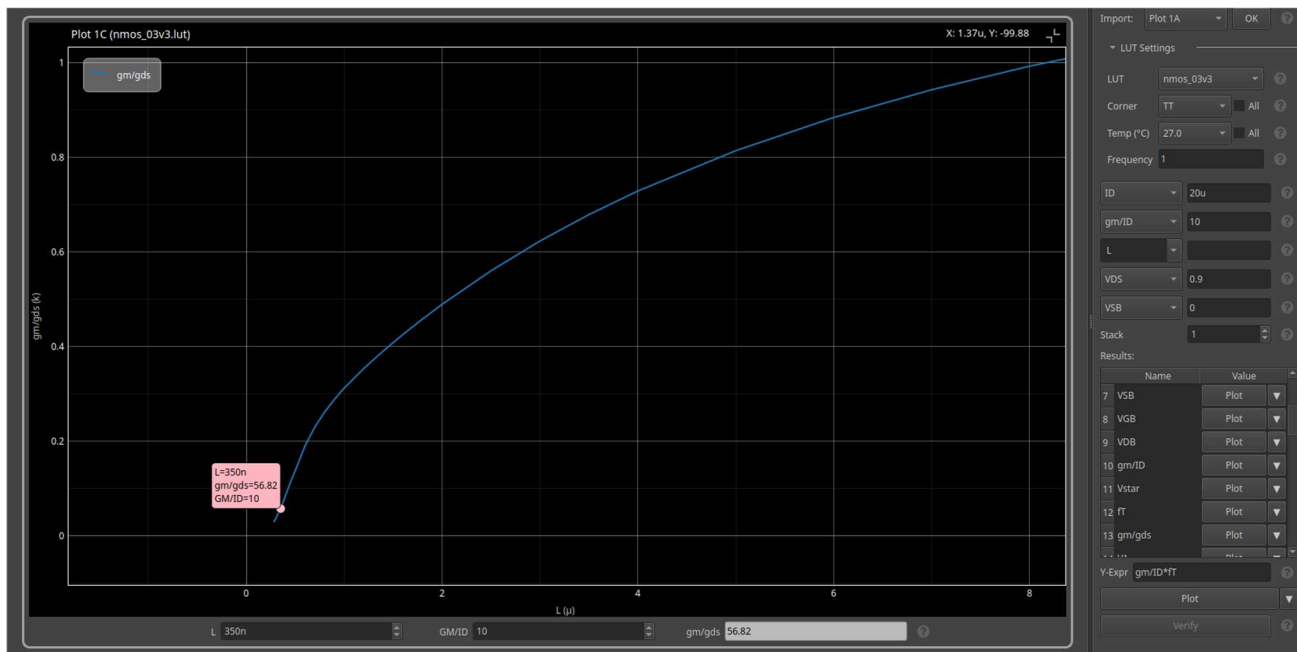
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DC OP Analysis

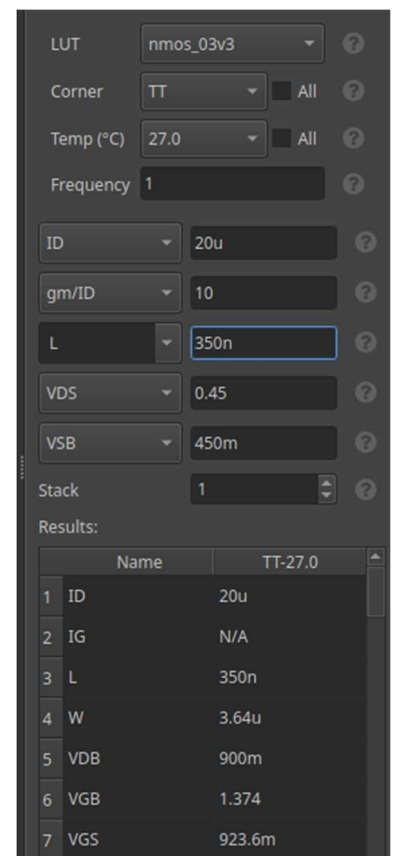
Sizing

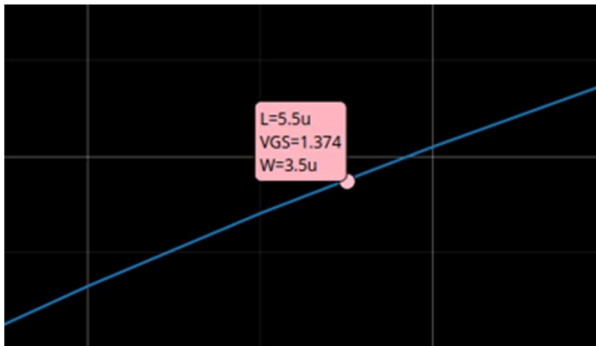
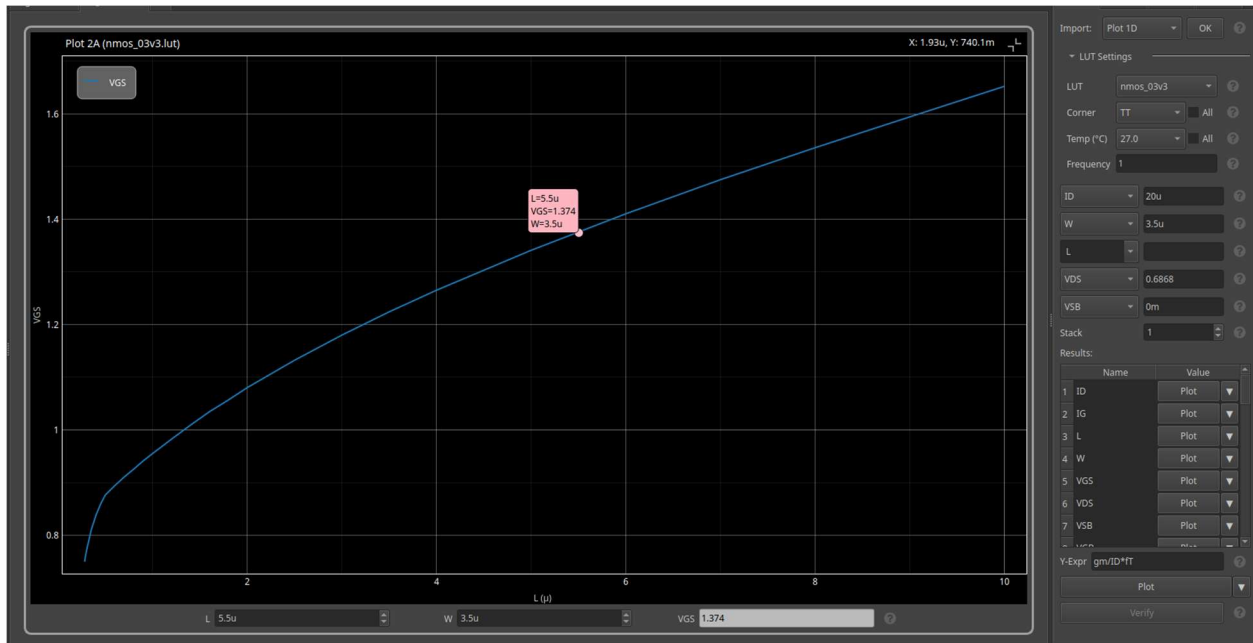
First we need to calculate the L of the nmos



After we got the L we needed to calculate VB to do so we will use the same sizing but change the Vds and vbs due to body affect and that we use a cascode so we assume the vout dc is distributed on each as equal as 450mv and get Vgs.

After we Vgs we can proceed by saying $V_B = V_{gs} + 0.45 = 1.3736$ the we need our magic battery to produce such voltage so we will increase the L





Rd Calculation

Total supply voltage: $V_{DD} = 1.8V$

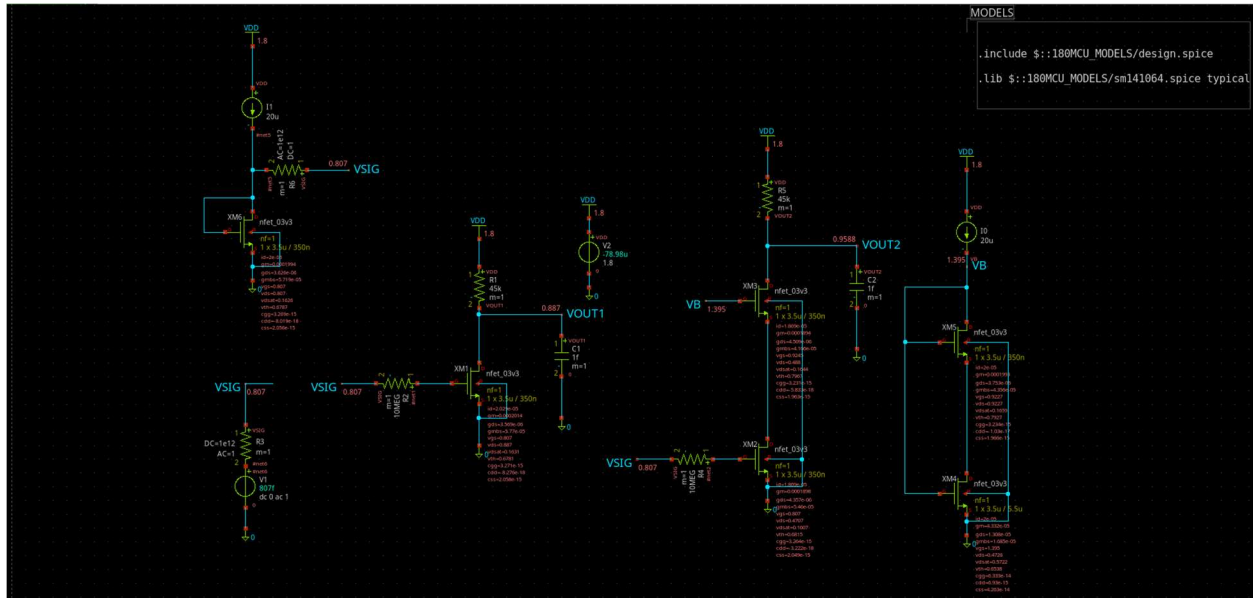
$$V_{drop} = \frac{V_{DD}}{2} = 0.9V, I_D \approx 20\mu A$$

$$R_D = \frac{V_{drop}}{I_D} = \frac{0.9}{20 \times 10^{-6}} = 45k\Omega$$

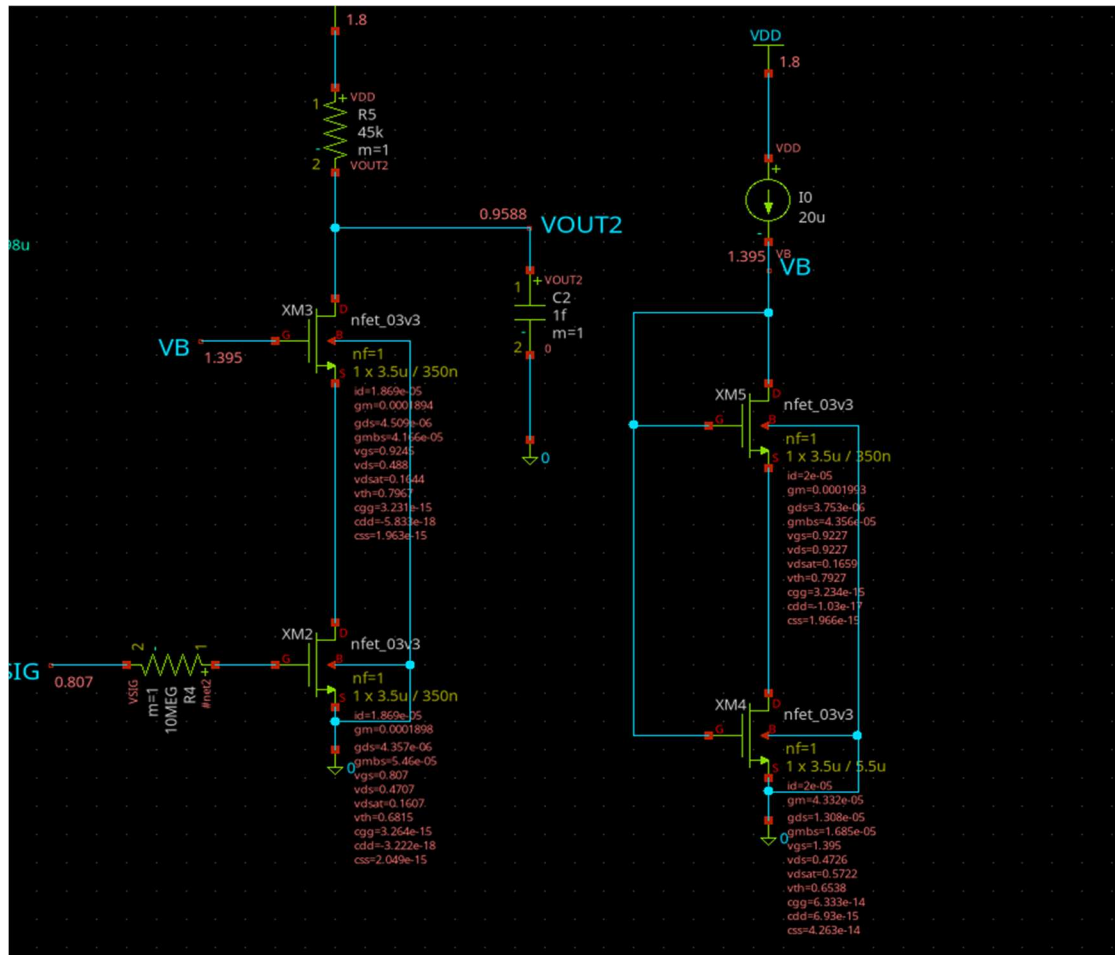


Schematic

main:

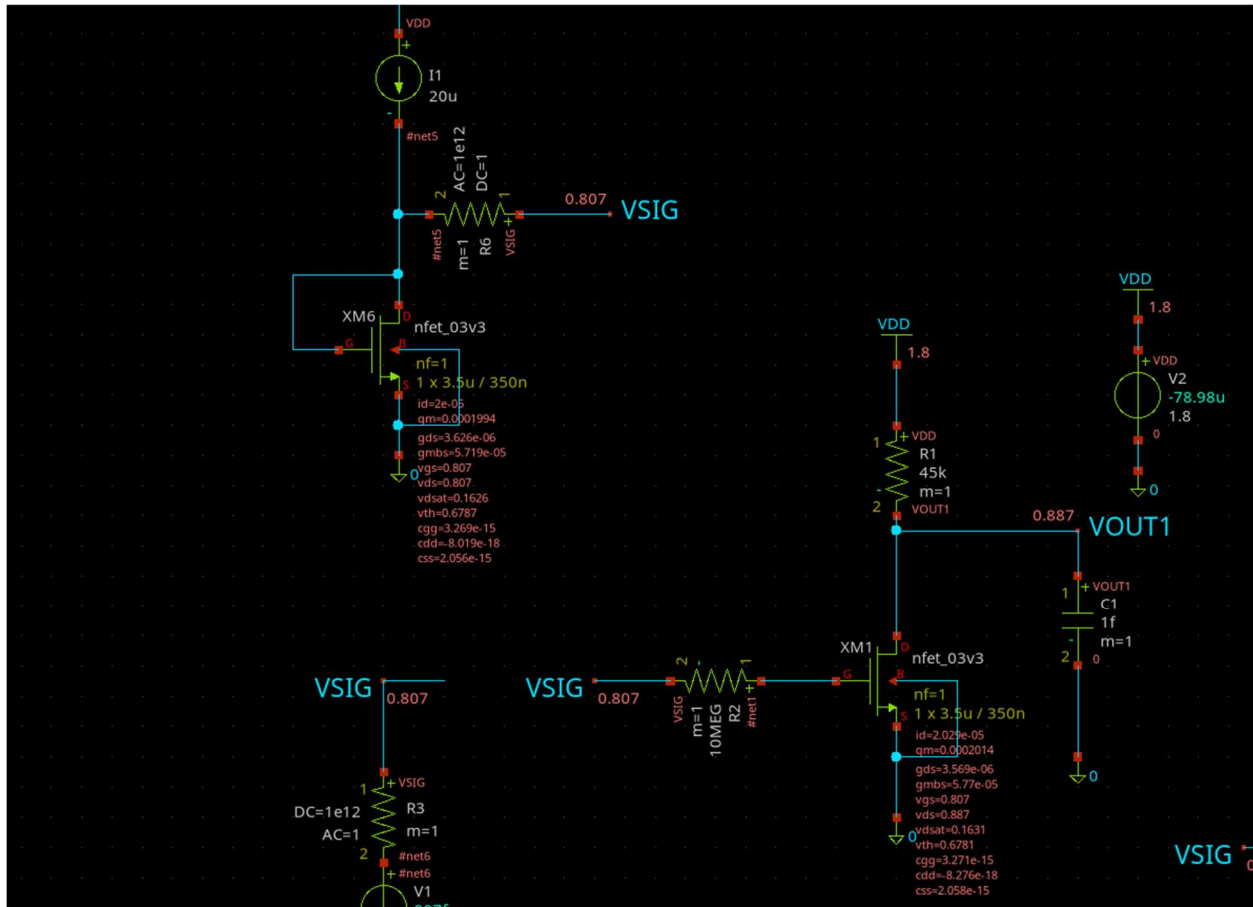


Cascode and magic battery:





Common source:



Saturation op check

M1, M2, M3, M5, and M6 operate in the saturation region because, for each transistor, $V_{DS} \geq V_{DSAT}$.

M4 is the only transistor operating in the triode region since $V_{DS} < V_{DSAT}$.

For M4,

$$V_{GD} = V_{GS} - V_{DS4}$$

$$\text{Since } V_{GS4} = V_{OV} + V_{TH} ,$$

$$V_{GD4} = V_{OV4} + V_{TH} - V_{DS4}$$

Because $V_{DS} < V_{OV4}$, then $V_{GD4} > V_{TH}$. Therefore, the channel is not pinched off, and M4 operates in the triode region.



AC Analysis

1) Perform AC analysis to simulate gain and bandwidth.

```
binary raw file "lab_03_cascode_for_BW.raw"
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

No. of Data Rows : 101
dc_gain_1      = 7.806966e+00
bw_1           = 1.694538e+06
ugf_1          = 1.318210e+07
dc_gain_2      = 8.355493e+00
bw_2           = 3.005948e+06
ugf_2          = 2.491143e+07
binary raw file "lab_03_cascode_for_newBW.raw"
binary raw file "/home/ubuntu/lab3/simulation/lab3_cascodeBW.sch/lab3_cascodeBW.
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver
OP information in rawfile.

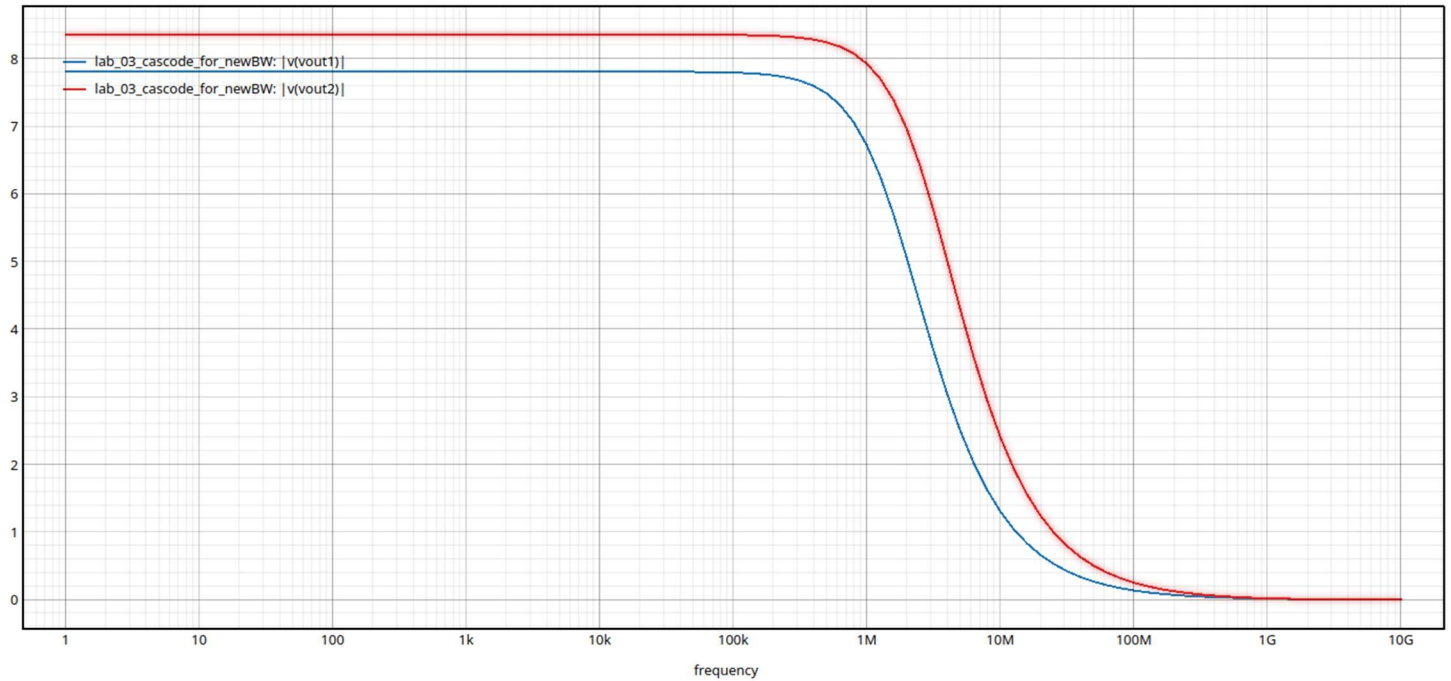
Total analysis time (seconds) = 0.00377663
```

2) Using measure to produce a txt file here are its content

```
dc_gain_1 = 7.806966e+00
bw_1 = 1.694538e+06
gbw_1 = 1.322920e+07
ugf_1 = 1.318210e+07
dc_gain_2 = 8.355493e+00
bw_2 = 3.005948e+06
gbw_2 = 2.511618e+07
ugf_2 = 2.491143e+07
```



3) Report the Bode plot of CS and cascode appended on the same plot



Vout 1:CS ,Vout 2:Cascode



4) Using small signal parameters from OP simulation or SA, perform hand analysis to calculate DC gain, BW, and GBW of both circuits

1. Common-Source Amplifier

$$g_{m1} = 201.4 \mu\text{S}, \quad g_{ds} = 3.569 \mu\text{S}$$

$$r_{o1} = \frac{1}{g_{ds}} = \frac{1}{3.569 \times 10^{-6}} = 280.2 \text{ k}\Omega$$

$$R_{out1} = R_D \parallel r_{o1} = 45 \text{ k}\Omega \parallel 280.2 \text{ k}\Omega = 38.77 \text{ k}\Omega$$

$$A_{v1} = -g_{m1}R_{out1} = -(201.4 \times 10^{-6})(38.77 \times 10^3) = -7.81 \text{ V/V}$$

$$C_{in1} \approx C_{gs1} + C_{gd1}(1 + |A_{v1}|) \approx 9.5 \text{ fF}$$

$$BW_1 \approx \frac{1}{2\pi R_G C_{in}}$$

$$BW_1 \approx \frac{1}{2\pi(10 \times 10^6)(9.5 \times 10^{-15})} = 1.68 \text{ MHz}$$

$$GBW_1 = |A_{v1}|BW_1 = (7.81)(1.68 \times 10^6) = 13.08 \text{ MHz}$$

2. Cascode Amplifier

$$g_{m2} = 189.8 \mu\text{S}, \quad g_{ds} = 4.357 \mu\text{S}$$

$$g_{m3} = 189.4 \mu\text{S}, \quad g_{mb} = 41.66 \mu\text{S}, \quad g_{ds} = 4.509 \mu\text{S}$$

$$r_{o2} = \frac{1}{g_{ds2}} = \frac{1}{4.357 \times 10^{-6}} = 229.5 \text{ k}\Omega$$

$$r_{o3} = \frac{1}{g_{ds}} = \frac{1}{4.509 \times 10^{-6}} = 221.8 \text{ k}\Omega$$

$$r_{o,cas} = r_{o2} + r_{o3} + (g_{m3} + g_{mb3})r_{o2}r_{o3}$$

$$r_{o,cas} = 229.5 \text{ k}\Omega + 221.8 \text{ k}\Omega + (189.4 + 41.66) \times 10^{-6}(229.5 \times 10^3)(221.8 \times 10^3)$$

$$r_{o,cas} \approx 12.21 \text{ M}\Omega$$

$$R_{out2} = R_D \parallel r_{o,cas} = 45 \text{ k}\Omega \parallel 12.21 \text{ M}\Omega = 44.83 \text{ k}\Omega$$

$$A_{v2} \approx -g_{m2}R_{out2} = -(189.8 \times 10^{-6})(44.83 \times 10^3) = -8.51 \text{ V/V}$$

$$C_{in} \approx C_{gs} + C_{gd2}(1 + |A_x|) \approx 5.4 \text{ fF}$$

$$BW_2 \approx \frac{1}{2\pi R_G C_{in}}$$

$$BW_2 \approx \frac{1}{2\pi(10 \times 10^6)(5.4 \times 10^{-15})} = 2.95 \text{ MHz}$$

$$GBW_2 = |A_{v2}|BW_2 = (8.51)(2.95 \times 10^6) = 25.08 \text{ MHz}$$



5) Report a table comparing the DC gain, BW, UGF, and GBW of both circuits from simulation and hand analysis.

Parameter	Common Source Hand	Common Source Simulation	Cascode Hand	Cascode Simulation
DC gain (V/V)	7.81	7.807	8.51	8.355
Bandwidth, BW (MHz)	1.68	1.695	2.95	3.006
Unity-gain frequency, UGF (MHz)	13.01	13.182	24.93	24.911
Gain-bandwidth product, GBW (MHz)	13.08	13.229	25.08	25.116

Comment:

Suppression of the Miller effect:

The simulation and hand calculations show that the cascode amplifier has a bandwidth of approximately 3.0 MHz, compared with 1.69 MHz for the common-source amplifier. This is an improvement of about 77%. In the cascode circuit, M_3 keeps the drain voltage of the input transistor M_2 from changing significantly. Therefore, C_{gd} experiences much less Miller multiplication, reducing the effective input capacitance and moving the input pole to a higher frequency.

DC gain:

The two amplifiers have similar DC gains: approximately 7.81 V/V for the common-source amplifier and 8.36 V/V for the cascode amplifier. Although the cascode provides a much larger intrinsic output resistance, the total output resistance is mainly limited by the 45 k Ω drain resistor. As a result, the increase in DC gain is relatively small.

Hand analysis versus simulation:

The hand-calculated results agree closely with the Ngspice simulation. The small differences, especially for the cascode gain and bandwidth, are expected because the hand analysis uses simplified first-order small-signal equations. Ngspice includes the exact operating point, body effect, short-channel effects, and the complete nonlinear parasitic-capacitance model.

GBW and UGF improvement:

The simulated GBW increases from 13.23 MHz for the common-source amplifier to 25.12 MHz for the cascode amplifier, which is an improvement of about 90%. The UGF shows a similar increase, from 13.18 MHz to 24.91 MHz. Therefore, the cascode provides a much better high-frequency response while maintaining approximately the same DC gain.

**Voltage-headroom trade-off:**

The improved frequency response comes at the cost of voltage headroom. Since the cascode uses two stacked transistors, enough voltage must be provided to keep both devices in saturation. This reduces the available output-voltage swing and makes the cascode less suitable for very low supply voltages compared with the simple common-source amplifier.